

Parameter Estimation

Theorem (M, Manole, Rigollet, forthcoming)

Observe $Y_i = f_\mu(X_i) + \epsilon_i$, $i = 1, \dots, n$ with $\epsilon_i \sim \mathcal{N}(0,1)$, $X_i \sim \mathcal{U}([0,1])$ independently. Set $Q_{f_\mu} = \text{Law}(X_1, Y_1)$. We define the minimax risk,

$$\mathcal{R}_n = \inf_{\hat{\mu}_n} \sup_{\mu \in \bar{G}_k} \mathbb{E} [W_1(\hat{\mu}_n, \mu)]$$

in expectation over $Q_{f_\mu}^{\otimes n}$. There exists $C_k > 0$ such that,

$$\frac{1}{C_k} \left(\frac{1}{n} \right)^{\frac{1}{2k}} \leq \mathcal{R}_n \leq C_k \left(\frac{\log^2(n)}{n} \right)^{\frac{1}{2k}}$$

Upper bound: achieved by square loss ERM for $f_\mu(\cdot)$.

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Intuition for lower bound: poor stability of recovering uniform atomic measures from moments.

Lemma (Hundrieser et al. 2025)

There exist $s, v \in \mathcal{A}_k$ which are distinct and have $\mathbb{E}_s[x^l] = \mathbb{E}_u[x^l]$ for $1 \leq l \leq k-1$.

For $s_n = (n^{-1/2k}(\cdot))_{\#}s$, u_n same, we have: $W_1(s_n, u_n) \gtrsim n^{-\frac{1}{2k}}$

Two-point testing reduction: $\chi^2(f_{s_n}, f_{u_n}) \leq \sum_{l=1}^{\infty} \frac{(\mathbb{E}_{s_n}[x^l] - \mathbb{E}_{u_n}[x^l])^2}{l!} \lesssim \frac{1}{n}$